

| Item Name                                                                                   | Qty | # | Price (total) (USD) | Item link                                                                                                                                                                                                                                                     | Notes                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------|-----|---|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Flywoo Goku F405 HD AIO V2                                                                  | 1   |   | 69.99               | <a href="https://flywoo.net/products/goku-f405-hd-1-2a-elfs-aio-v2">https://flywoo.net/products/goku-f405-hd-1-2a-elfs-aio-v2</a>                                                                                                                             | Drone flight controller                                                                                                                                                                       |
| T-Motor F1404 (4600KV)                                                                      | 4   |   | 78.8                | <a href="https://robokit5.co.in/multitrotor-spare-parts/tiger-motor-esc-and-propeller/t-motor-motoroff-type/t-motor-f1404-kv4600">https://robokit5.co.in/multitrotor-spare-parts/tiger-motor-esc-and-propeller/t-motor-motoroff-type/t-motor-f1404-kv4600</a> | Drone motor, earlier i thought of using different motor(RCinPower 1303 11500kv) but it was simply not powerful enough to lift whole drone neither the whole drone body itself                 |
| TBS Unify Pro32 Nano VTX                                                                    | 1   |   | 35.44               | <a href="https://www.quadkart.in/tbs-unify-pro32-nano-5g8-v1-1/">https://www.quadkart.in/tbs-unify-pro32-nano-5g8-v1-1/</a>                                                                                                                                   | For video transferring from drone camera to ground unit(i.e. pilot)                                                                                                                           |
| HGLRC M100 Mini GPS                                                                         | 1   |   | 16.67               | <a href="https://www.quadkart.in/hglrc-m100-mini-gps/">https://www.quadkart.in/hglrc-m100-mini-gps/</a>                                                                                                                                                       | i used this one simply because of its size                                                                                                                                                    |
| RunCam Split 4 V3                                                                           | 1   |   | 89                  | <a href="https://shop.runcam.com/runcam-split-4-v3/">https://shop.runcam.com/runcam-split-4-v3/</a>                                                                                                                                                           | i used this one because its great for post flight analysis of the environment i.e. when the measurement were taken and during the flight for video quality                                    |
| Seed Studio XIAO ESP32-S3 Sense                                                             | 1   |   | 13.5                | <a href="https://in.element14.com/seeed-studio/113991115/development-board-32bit-xtensa/dp/4200197">https://in.element14.com/seeed-studio/113991115/development-board-32bit-xtensa/dp/4200197</a>                                                             | Pod brain and data logger with its board                                                                                                                                                      |
| 7Semi BME688 Nano Breakout with pitch 2.54mm                                                | 1   |   | 10.52               | <a href="https://evelta.com/7semi-bme688-environmental-sensor-nano-breakout-2-54mm-pitch-header/">https://evelta.com/7semi-bme688-environmental-sensor-nano-breakout-2-54mm-pitch-header/</a>                                                                 | i used this one due to its size and i made the pod 2nd enclosure based on this size, its pitch distance is 2.54mm                                                                             |
| SparkFun Air Quality BMV080                                                                 | 1   |   | 59.95               | <a href="https://www.sparkfun.com/products/26554">https://www.sparkfun.com/products/26554</a>                                                                                                                                                                 | this sensor uses laser to measure the pollutants in air upto PM10(which is great tbh)                                                                                                         |
| Flywoo Explorer 2S 18650 Pack                                                               | 1   |   | 20                  | <a href="https://flywoo.net/products/flywoo-explorer-molicell-p30b-18650-2s1p-3000mah-lionpack-battery">https://flywoo.net/products/flywoo-explorer-molicell-p30b-18650-2s1p-3000mah-lionpack-battery</a>                                                     | its a 2S 18650 battery pack for more power delivery and better flight time                                                                                                                    |
| XT30UP Power Cable & Filter Cap                                                             | 1   |   | 1                   | <a href="https://robu.in/product/safeconnect-xt-30-male-16awg-20cm-connector-pigtail-cable/">https://robu.in/product/safeconnect-xt-30-male-16awg-20cm-connector-pigtail-cable/</a>                                                                           | to connect the power and to power the pcb of pod                                                                                                                                              |
| Custom PCB                                                                                  | 1   |   | 12.75               | <a href="https://github.com/adityaanand7982-droid/Project-Mote/tree/main/PCB">https://github.com/adityaanand7982-droid/Project-Mote/tree/main/PCB</a>                                                                                                         | i made pcb to keep everything neat and clean and keep the size as small as possible with neatly fitting in the pod                                                                            |
| Custom 3D printable files                                                                   | 1   |   | 15                  | <a href="https://github.com/adityaanand7982-droid/Project-Mote/tree/main/CAD">https://github.com/adityaanand7982-droid/Project-Mote/tree/main/CAD</a>                                                                                                         | it includes printable parts(pod, pod cover, camera enclosure, battery enclosure)                                                                                                              |
| Drone Frame (3mm FR4)                                                                       | 1   |   | 30.6                | <a href="https://github.com/adityaanand7982-droid/Project-Mote/tree/main/CAD/Part%201(Frame)">https://github.com/adityaanand7982-droid/Project-Mote/tree/main/CAD/Part%201(Frame)</a>                                                                         | i wasn't able to find the carbon fibre sheet cnc service neither on any websites or locals so after few researchs i decided to go with FR4 which can be easily sourced from jlcpbc and pcbway |
| Gemfan 3018 Props bi blade                                                                  | 1   |   | 3.74                | <a href="https://www.uavmarketplace.in/products/gemfan-3018-hurricane-pc-414r-2-blade-propeller/89578900000939622">https://www.uavmarketplace.in/products/gemfan-3018-hurricane-pc-414r-2-blade-propeller/89578900000939622</a>                               | Standard 8-Pack (4 CW + 4 CCW) for flying, now the reason i went with 2 blades rather than 3 blades, coz 2 blades simply are more energy efficient than 3 blades                              |
| TrueRC Singularity 5.8GHz Ant( UFL / RHCP)                                                  | 1   |   | 20                  | <a href="https://truerc.com/71966-72167-singularity-58-ufll/753-polarization-rhcp/449-connector-cable-ufll-regular-72mm">https://truerc.com/71966-72167-singularity-58-ufll/753-polarization-rhcp/449-connector-cable-ufll-regular-72mm</a>                   | Norma video antenna for drone(MUST BE SAME AS RECEIVER, here i choose RHCP)                                                                                                                   |
| Metric Fastener Array Kit                                                                   | 1   |   | 12                  | <a href="https://www.amazon.in/Standoff-Mounting-Screws-GigaByte-Motherboards/dp/B0DG2873WB">https://www.amazon.in/Standoff-Mounting-Screws-GigaByte-Motherboards/dp/B0DG2873WB</a>                                                                           | miscellinous but important stuffs like mountings the flight controllers                                                                                                                       |
| 1N5819 Schottky Diode                                                                       | 1   |   | 0.03                | <a href="https://robu.in/product/1n5819-1w-diode-pack-of-30">https://robu.in/product/1n5819-1w-diode-pack-of-30</a>                                                                                                                                           | it prevents backward flow of current from esp to drone's battery if plugged into esp32 for post flight data analysis                                                                          |
| Rubber standoffs                                                                            | 2   | 1 |                     | <a href="https://evelta.com/4pc-m2-anti-vibration-rubber-shock-absorber-ball/">https://evelta.com/4pc-m2-anti-vibration-rubber-shock-absorber-ball/</a>                                                                                                       | For vibration damping                                                                                                                                                                         |
| Threadlocker                                                                                | 1   |   | 8.32                | <a href="https://www.amazon.in/Permatex-24010-Strength-Threadlocker-Applicator/dp/B000HBM564">https://www.amazon.in/Permatex-24010-Strength-Threadlocker-Applicator/dp/B000HBM564</a>                                                                         | For locking the screws, so it won't fall out from vibrations.                                                                                                                                 |
| Battery Strap                                                                               | 1   |   | 0.6                 | <a href="https://robu.in/product/20cm-lipo-battery-strap-belt-reusable-cable-tie-wrap/">https://robu.in/product/20cm-lipo-battery-strap-belt-reusable-cable-tie-wrap/</a>                                                                                     | For holding the battery                                                                                                                                                                       |
| TOTAL                                                                                       | 20  |   | 498.9               | 69.99 + 78.80 + 35.44 + 16.67 + 89.00 + 13.50 + 10.52 + 59.95 + 20.00 + 1.00 + 12.75 + 15.00 + 30.60 + 3.74 + 20.00 + 12.00 + 0.03 + 1.00 + 8.32 + 0.60                                                                                                       |                                                                                                                                                                                               |
| NOTE: All the 3d printed parts + frame CNC(FR4)+ PCB total came out to be around \$60 total |     |   |                     |                                                                                                                                                                                                                                                               |                                                                                                                                                                                               |